Exercise Sheet

Lecture 8

Orthonormal Bases, Gram–Schmidt, and Riesz

The inner product is taken in the physics convention: conjugate-linear in the first slot, linear in the second. Unless stated otherwise, the inner product is the standard one. Tags mark the flavour; a ⋆ marks a harder one.

Core tutorial route

Work **8.1(i), 8.4(i), 8.4(iii), 8.5(i), 8.7(i), 8.9(i), and 8.12** in that order. This route covers coordinates in an orthonormal basis and Parseval's relation, real and complex Gram–Schmidt, an orthogonal complement, best approximation, least squares, and Riesz for a non-standard inner product.

Additional practice: 8.1(ii), 8.2, 8.4(ii), 8.5(ii), 8.6, 8.7(ii)–(iii), 8.8, 8.9(ii), 8.11(i)–(ii). **Bridge:** 8.3, 8.10, 8.11(iii) (the matrix of an operator, polynomial and matrix spaces). **Extension:** 8.13 (the kernel of a functional).

Orthonormal bases

Exercise 8.1 (computational) Expand the vector v in the given orthonormal basis and verify Parseval's relation, for (i) $v = (2,3) \in \mathbb{R}^2$, $e_1 = \frac{1}{\sqrt{5}}(1,2)$, $e_2 = \frac{1}{\sqrt{5}}(2,-1)$; (ii) $v = (3,0,0) \in \mathbb{R}^3$, $e_1 = \frac{1}{3}(1,2,2)$, $e_2 = \frac{1}{3}(2,1,-2)$, $e_3 = \frac{1}{3}(2,-2,1)$.

Answer. (i) $v = \frac{8}{\sqrt{5}}e_1 + \frac{1}{\sqrt{5}}e_2$; Parseval's relation: $\frac{64}{5} + \frac{1}{5} = 13 = \|v\|^2$. (ii) $v = e_1 + 2e_2 + 2e_3$; Parseval's relation: $1 + 4 + 4 = 9 = \|v\|^2$.

Exercise 8.2 (proof)

(i) Let the pairwise orthogonal (not necessarily unit) vectors $v_1, \dots, v_n$ form a basis of V. Prove that for every $v \in V$

$$v = \sum_k \frac{\langle v_k, v\rangle}{\|v_k\|^2}\, v_k.$$

(ii) Let $e_1, \dots, e_n$ be unit vectors in V such that $\|v\|^2 = \sum_{k=1}^{n} |\langle e_k, v\rangle|^2$ for every $v \in V$. Prove that the vectors $e_1, \dots, e_n$ form an orthonormal basis of V.

Hint. (i) Normalize the vectors, $e_k = v_k/\|v_k\|$, and expand v in the orthonormal basis $e_1, \dots, e_n$. (ii) For orthogonality, take $v = e_j$. To show that the vectors span V, apply the hypothesis to $r = v - \sum_k \langle e_k, v\rangle e_k$.

Exercise 8.3 (⋆) Let $e_1, \dots, e_n$ be an orthonormal basis of V and $T \in \mathcal{L}(V)$. (i) Explain why the matrix of T in this basis has entries $T_{jk} = \langle e_j, Te_k\rangle$. (ii) Prove that

$$\sum_{j,k} |T_{jk}|^2 = \sum_k \|Te_k\|^2,$$

and deduce that this sum is 0 only for $T = 0$.

Hint. (i) Column k of the matrix holds the coordinates of Te_k. (ii) Apply Parseval's relation to each vector Te_k.

The Gram–Schmidt process

Exercise 8.4 (computational) Apply the Gram–Schmidt process to (i) $v_1 = (3,4)$, $v_2 = (1,0)$ in $\mathbb{R}^2$; (ii) $v_1 = (1,1,1)$, $v_2 = (1,1,0)$, $v_3 = (1,0,0)$ in $\mathbb{R}^3$; (iii) $v_1 = (1, 1+i)$, $v_2 = (0,1)$ in $\mathbb{C}^2$.

Answer. (i) $e_1 = \frac{1}{5}(3,4)$, $e_2 = \frac{1}{5}(4,-3)$. (ii) $e_1 = \frac{1}{\sqrt{3}}(1,1,1)$, $e_2 = \frac{1}{\sqrt{6}}(1,1,-2)$, $e_3 = \frac{1}{\sqrt{2}}(1,-1,0)$. (iii) $e_1 = \frac{1}{\sqrt{3}}(1, 1+i)$, $e_2 = \frac{1}{\sqrt{3}}(-1+i, 1)$.

Hint. (iii) Over $\mathbb{C}$, do not forget to conjugate the coordinates of the first vector.

Orthogonal projection and best approximation

Exercise 8.5 (computational) For each subspace U below, find a basis of the orthogonal complement $U^\perp$ and check that $\dim U + \dim U^\perp = \dim V$: (i) $U = \operatorname{span}\big((1,0,1)\big) \subset \mathbb{R}^3$; (ii) $U = \operatorname{span}\big((1,0,1,0),(0,1,0,1)\big) \subset \mathbb{R}^4$.

Answer. (i) $U^\perp = \operatorname{span}\big((0,1,0),(-1,0,1)\big)$, $\dim U^\perp = 2$; $1+2=3$. (ii) $U^\perp = \operatorname{span}\big((-1,0,1,0),(0,-1,0,1)\big)$, $\dim U^\perp = 2$; $2+2=4$.

Exercise 8.6 (proof) Let U and W be subspaces of V. Prove that (i) $U \subseteq W$ implies $W^\perp \subseteq U^\perp$; (ii) $(U+W)^\perp = U^\perp \cap W^\perp$.

Hint. Work from the definition of the orthogonal complement. To prove the inclusion $\supseteq$ in (ii), expand $\langle u+w, v\rangle$.

Exercise 8.7 (computational) Find the point $P_U b$ of U closest to the vector b, and the distance from b to U, for (i) $U = \operatorname{span}\big((1,2,-2)\big) \subset \mathbb{R}^3$, $b = (0,3,0)$; (ii) $U = \operatorname{span}\big((1,1)\big) \subset \mathbb{R}^2$, $b = (2,4)$; (iii) $U \subset \mathbb{R}^3$ the plane with orthonormal basis $e_1 = \frac{1}{\sqrt{2}}(1,1,0)$, $e_2 = \frac{1}{\sqrt{6}}(1,-1,2)$, and $b = (3,0,0)$.

Answer. (i) $P_U b = \frac{2}{3}(1,2,-2)$; distance $\sqrt{5}$. (ii) $P_U b = (3,3)$; distance $\sqrt{2}$. (iii) $P_U b = (2,1,1)$; distance $\sqrt{3}$.

Exercise 8.8 (proof) Let P_U be the orthogonal projection onto a subspace U with orthonormal basis $e_1, \dots, e_m$. Prove that (i) $P_U^2 = P_U$; (ii) $\langle P_U v, w\rangle = \langle v, P_U w\rangle$ for all $v, w \in V$.

Hint. (i) $P_U v$ lies in U; what does P_U do to a vector of U? (ii) Expand $P_U v$ on the left and $P_U w$ on the right, and compare the two sums term by term, watching where the conjugation falls.

Exercise 8.9 (computational) Use least squares to (i) fit a line $y = c_0 + c_1 x$ to the points $(1,1)$, $(2,2)$, $(3,2)$; (ii) solve the inconsistent system $Ax = b$ with $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$ and $b = (1,1,0)$.

Answer. (i) $y = \frac{2}{3} + \frac{1}{2}x$. (ii) $\hat{x} = \left(\frac{1}{3}, \frac{1}{3}\right)$.

Exercise 8.10 ($\star$) On $\mathcal{P}_3$ with the inner product $\langle p, q\rangle = \int_{-1}^{1} \overline{p}\, q \, dx$, find the polynomial of degree at most one closest to $f(x) = x^3$. Verify that the error is orthogonal to $\mathcal{P}_1$.

Hint. Project x^3 onto $\mathcal{P}_1 = \operatorname{span}(1, x)$, using the lecture's orthonormal polynomials $p_0 = \frac{1}{\sqrt{2}}$ and $p_1 = \sqrt{\frac{3}{2}}\, x$. An odd function integrates to 0 over $[-1,1]$.

Answer. The closest polynomial is $\frac{3}{5}x$; the error is $x^3 - \frac{3}{5}x$.

The Riesz representation theorem

Exercise 8.11 (computational) Find the vector w such that $\varphi(v) = \langle w, v\rangle$, for (i) $\varphi(v) = 2v_1 - iv_2$ on $\mathbb{C}^2$; (ii) $\varphi(v) = 2v_1 - v_2 + 3v_3$ on $\mathbb{R}^3$; (iii) $\varphi(A) = 2A_{11} - A_{12} + A_{22}$ on $M_{2\times 2}(\mathbb{R})$ with the Frobenius product $\langle A, B\rangle = \operatorname{tr}(A^\dagger B)$ (here w is a matrix).

Answer. (i) $w = (2, i)$. (ii) $w = (2, -1, 3)$. (iii) $w = \left(\begin{smallmatrix} 2 & -1 \\ 0 & 1 \end{smallmatrix}\right)$.

Exercise 8.12 (computational) On $\mathbb{C}^2$ take the weighted inner product $\langle u, v\rangle_2 = \overline{u_1}v_1 + 2\,\overline{u_2}v_2$ of Lecture 7. Find the vector w such that $\varphi(v) = \langle w, v\rangle_2$, where $\varphi(v) = 2v_1 - iv_2$ is the functional of Exercise 8.11(i). Compare with the answer there.

Hint. As in Exercise 8.11(i), match the coefficients of v_1 and v_2; only now v_2 carries the weight 2.

Answer. $w = \left(2, \frac{i}{2}\right)$; for the standard product it was $(2, i)$, so the vector that represents a functional depends on the inner product.

Exercise 8.13 ($\star$) Let V be a finite-dimensional space and φ a linear functional on V. By Riesz there is a vector $w \in V$ such that $\varphi(v) = \langle w, v\rangle$ for all $v \in V$. Prove that $\ker\varphi = (\operatorname{span}(w))^\perp$, and that $\dim\ker\varphi = \dim V - 1$ when $w \neq 0$.

Hint. Read $\varphi(v) = 0$ as an orthogonality condition.